

Differentiation by first principles



1

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	37
3	3.5	0.5	31.75
3	3.1	0.1	27.91
3	3.05	0.05	27.4525
3	3.01	0.01	27.0901
3	3.001	0.001	27.009
3	3.0001	0.0001	27.0009

b

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	3	12	27	48	75

c $3x^2$

2

a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	7
3	3.5	0.5	6.5
3	3.1	0.1	6.1
3	3.05	0.05	6.05
3	3.01	0.01	6.01
3	3.001	0.001	6.001
3	3.0001	0.0001	6.0001

b

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	2	4	6	8	10

c $2x$

3



a

a	b	$h = b - a$	$\frac{f(b) - f(a)}{b - a}$
3	4	1	14
3	3.5	0.5	13
3	3.1	0.1	12.2
3	3.05	0.05	12.1
3	3.01	0.01	12.02
3	3.001	0.001	12.002
3	3.0001	0.0001	12.0002

b

x	1	2	3	4	5
Instantaneous rate of change of $f(x)$ at x	4	8	12	16	20

c $4x$

4

a

i $3x + 3h + 2$ ii $3h$ iii 3

b

i $8 - x - h$ ii $-h$ iii -1

c

i $2x + 2h - 3$ ii $2h$ iii 2

d

i $-5x - 5h + 4$ ii $-5h$
iii -5

e

i $5 - x^2 - 2hx - h^2$ ii $-2hx - h^2$
iii $-(2x + h)$

f

i $x^2 + 2hx + h^2 + 3x + 3h + 6$ ii $2hx + h^2 + 3h$
iii $2x + h + 3$

g

i $\frac{1}{x + h}$ ii $-\frac{h}{x(x + h)}$ iii $-\frac{1}{x(x + h)}$

h

i $\frac{1}{7(x + h)}$ ii $-\frac{h}{7x(x + h)}$ iii $-\frac{1}{7x(x + h)}$

i

$$\text{i} \quad 3x^2 + 6hx + 3h^2 - 4x - 4h + 2$$

$$\text{iii} \quad 6x + 3h - 4$$



$$\text{ii} \quad 6hx + 3h^2 - 4h$$

5

a

$$\text{i} \quad f(x+h) = x^2 + 2hx + h^2$$

$$\text{iii} \quad \frac{f(x+h) - f(x)}{h} = 2x + h$$

$$\text{ii} \quad f(x+h) - f(x) = 2hx + h^2$$

$$\text{iv} \quad \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) = 2x$$

b

$$\text{i} \quad f(x+h) = 4x^2 + 8hx + 4h^2$$

$$\text{iii} \quad \frac{f(x+h) - f(x)}{h} = 8x + 4h$$

$$\text{ii} \quad f(x+h) - f(x) = 8hx + 4h^2$$

$$\text{iv} \quad \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) = 8x$$

c

$$\text{i} \quad f(x+h) = -6x^2 - 12hx - 6h^2$$

$$\text{iii} \quad \frac{f(x+h) - f(x)}{h} = -12x - 6h$$

$$\text{ii} \quad f(x+h) - f(x) = -12hx - 6h^2$$

$$\text{iv} \quad \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right) = -12x$$

6

$$\text{a} \quad 0$$

$$\text{b} \quad 0$$

$$\text{c} \quad 5$$

$$\text{d} \quad -2$$

$$\text{e} \quad 2$$

$$\text{f} \quad 2x$$

$$\text{g} \quad 8x$$

$$\text{h} \quad 2x - 3$$

$$\text{i} \quad 4x - 3$$

$$\text{j} \quad 8x - 3$$

$$\text{k} \quad \frac{1}{2\sqrt{x}}$$

$$\text{l} \quad \frac{1}{2\sqrt{x-5}}$$

Gradients

7

$$\text{a} \quad \text{constant}$$

$$\text{b} \quad 4$$

$$\text{c} \quad 4$$

8

$$\text{a} \quad 2x + 8$$

$$\text{b} \quad 14$$

9

$$\text{a} \quad \frac{1}{2\sqrt{x}}$$

$$\text{b} \quad \frac{1}{10}$$

10

$$\text{a} \quad -2x$$

$$\text{b} \quad 2$$

11

a $2x + 3$



b 11

12

a $3x^2$

b 12

13

a $4x^3$

b -500

14 24

15 144

16 -4

17 60

18

a $-\frac{1}{x^2}$

b $x = \pm 4$

19

a $-\frac{1}{2(x-2)^2}$

b $-\frac{1}{8}$